

JeongHyeon Kim

M.S. Student, Mechanical Engineering, Korea University
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RESEARCH INTERESTS

My research interests lie in **control-oriented dynamics modeling** and **whole-body control**, with a focus on identifying task-relevant physical relationships and incorporating them into robust robot control.

EDUCATION

- **Korea University (GPA: 3.78/4.0, 4.25/4.5)** Mar. 2025 – Feb. 2027 (Expected)
M.S. in Mechanical Engineering Seoul, Republic of Korea
- **Korea University (GPA: 3.48/4.0, 3.9/4.5)** 2019 – 2025
B.S. in Mechanical Engineering Seoul, Republic of Korea

RESEARCH EXPERIENCE

- **Human-Machine Systems Lab, Korea University** Mar. 2024 – Present
Research Student Seoul, Republic of Korea
 - **Whole-body Adaptation for Humanoid Standing Manipulation**
 - Formulated a **Balance Demand Descriptor (BDD)** based on centroidal dynamics to represent the effects of upper-body motion, payload, and external interaction on whole-body balance.
 - Integrated BDD into locomotion control to explicitly account for changing balance demands during standing manipulation.
 - **Visual Perception for Quadrupedal Object Interaction**
 - Designed a dual-camera perception pipeline combining global workspace observation and local near-field sensing in a common coordinate frame.
 - Enabled real-time, task-specific perception under viewpoint changes and occlusion during locomotion.
 - **Real-time Architecture for a Generative-AI-based Mobile Manipulator**
 - Designed the real-time communication framework connecting simulation, an LLM, the user interface, and the physical robot.
 - Reorganized data flow around the robot's control cycle to resolve cross-module latency and meet real-time constraints.

PUBLICATIONS & MANUSCRIPTS

Manuscripts in Preparation

- **J. Kim** and S. Park, "Balance Demand Descriptor: A Physics-Informed Representation of Whole-Body Dynamics for Humanoid Standing Manipulation," *manuscript in preparation*.

Conference Presentations

- **J. Kim** and S. Park, "Obstacle Detection Using ORB-SLAM and UV-Disparity with a Low-Cost Quadruped Robot," *KSPE 2025 Spring Conference*, Jeju, Republic of Korea, May 2025.

SELECTED PROJECTS

- **Quadruped Robot Obstacle Avoidance (CCP) — Team Leader** Mar. 2024 – Jan. 2025
Computer Vision, Autonomous Navigation
 - Developed an obstacle detection and avoidance pipeline for the DOGZILLA quadruped robot.
 - Implemented ORB-SLAM for localization and mapping and UV-disparity for obstacle detection.
- **Ladder-Crossing Robot — Team Leader** Mar. 2023 – Sep. 2023

- Designed a ladder-crossing mechanism and implemented an Arduino-based control system.
- **Solar Tracking Parasol — Team Leader** *Sep. 2023 – Nov. 2023*
 - Designed the linkage mechanism and implemented a sun-tracking control system.

TEACHING EXPERIENCE

Teaching Assistant, Korea University, Seoul, Republic of Korea

- **Robotics** *Spring 2026*
- **Computer-Aided Design** *Fall 2025*
- **Dynamics (English)** *Summer 2025*
- **Kinematics of Mechanisms** *Spring 2025*

HONORS AND AWARDS

Fellowships and Scholarships

- **Master's Research Encouragement Grant**, National Research Foundation of Korea *2025*
Awarded to selected research-track graduate students for research funding support

Awards

- **President's Award**, Creative Challenger Program (CCP), Korea University *Jan. 2025*
Awarded for outstanding performance in autonomous obstacle avoidance for a quadruped robot
- **Bronze Award**, Creative Comprehensive Design Competition, Korea University *Sep. 2023*
- **Design Award**, Capstone Design Fair, Korea University *Nov. 2023*

TECHNICAL SKILLS

Programming: Python, C/C++, MATLAB
Robotics & Simulation: ROS2, Gazebo, Isaac Gym, MuJoCo, Pinocchio, URDF/XACRO
Learning & Optimization: Reinforcement Learning, Trajectory Optimization
Perception: OpenCV
Engineering & Development: Git, Arduino, Creo, NX, SolidWorks
OS: Linux (Ubuntu)